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International Decade of Sciences for Sustainable Development

STATUS AND TRENDS OF WOMEN IN SCIENCE

New Insights and Sectoral Perspectives

2025

This factsheet provides an update on the findings from the UNESCO publication:* The Gender Gap in Science: Status and Trends *(2024), offering a more detailed picture of where women are training and working in science. It draws on disaggregated data across sectors—public, private, higher education and non-profit. It also highlights women in quantum and water sciences, as special themes for 2025.

This UNESCO Factsheet is in support of the Call to Action to Close the Gender Gap in Science

June 2025

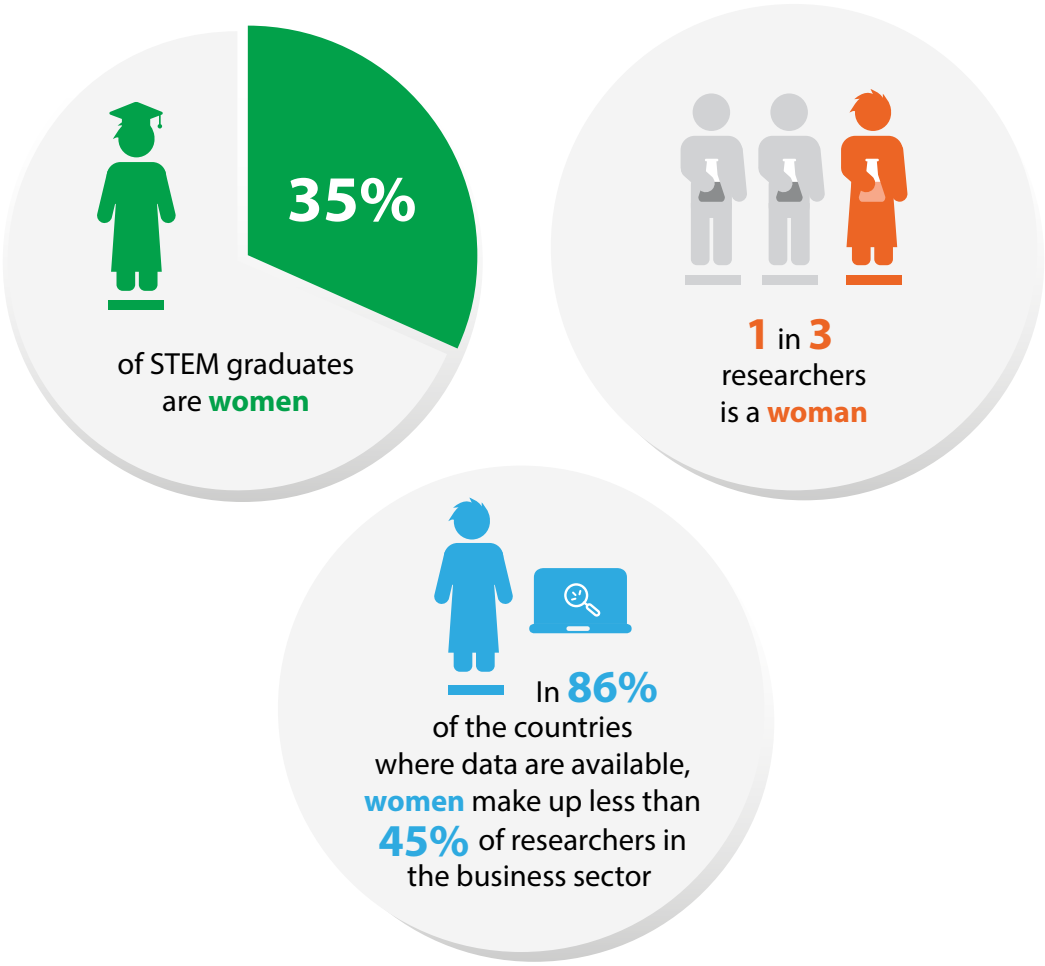
Gaps at a glance

Data series from UNESCO Member States coordinated by the UNESCO Institute for Statistics show three major gender gaps:

1 **Globally, young women are more likely than young men to enroll in higher education, with 46% of women and 40% of men enrolling in advanced studies within five years of completing their secondary education. But women account for 35% of science graduates.**

2 **Women form a persistently low share of scientific researchers. Just one in three researchers globally is a woman.**

3 **Women are moving towards equitable representation in the research workforce in academia and the public sector while men continue to dominate in the private sector, in the majority of countries.**

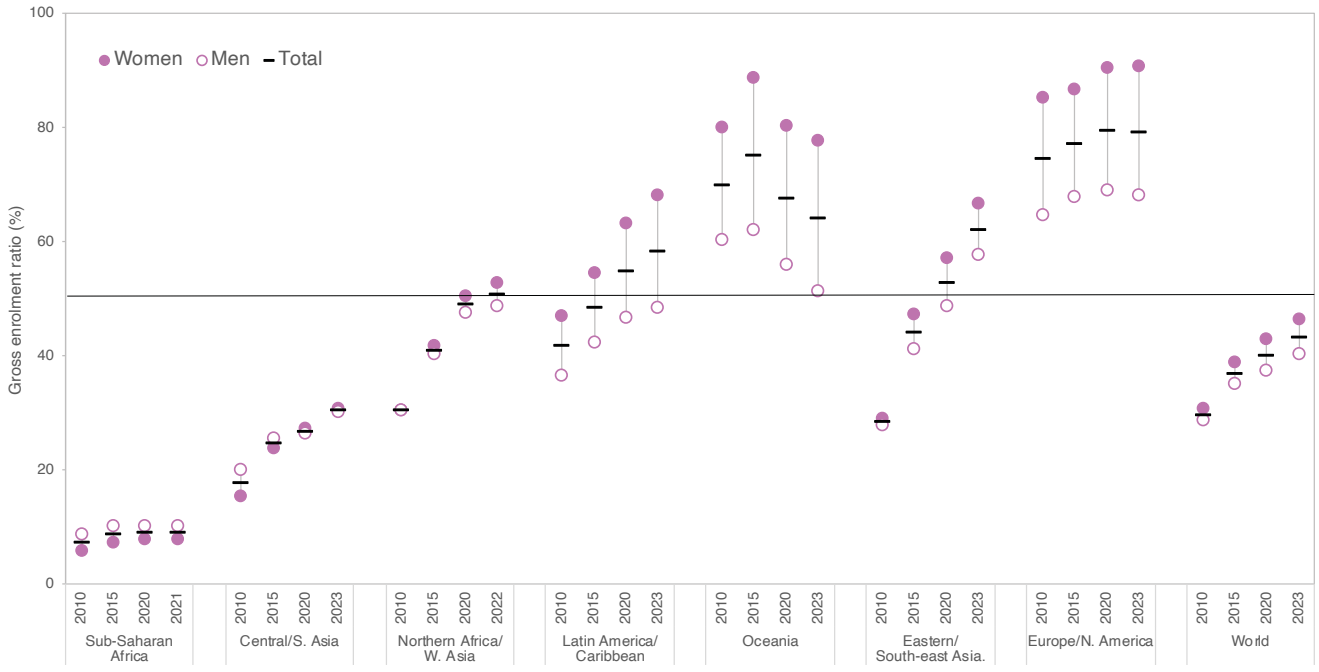


Source: UNESCO Institute for Statistics. Data from 2023 or most recent year; graduates data from 2018–2023. Business sector: based on 95 countries with data.

1 WOMEN ARE ACHIEVING IN HIGHER EDUCATION

Participation in tertiary education has been on the rise around the world. As described by UNESCO (2024a), "In most countries with available data, women considerably outnumber men in higher education (Figure 1), with 46% of women and 40% of men enrolling in advanced studies within five years of completing their secondary education. Across all regions, gender gaps in enrolment remained stable during the period 2010 to 2022. In most regions, except for sub-Saharan Africa, the gap is in favour of women. In Europe and Northern America, Latin America and the Caribbean, and Oceania, the gross enrolment ratio for women is at least 20 percentage points higher than that for men." [1]

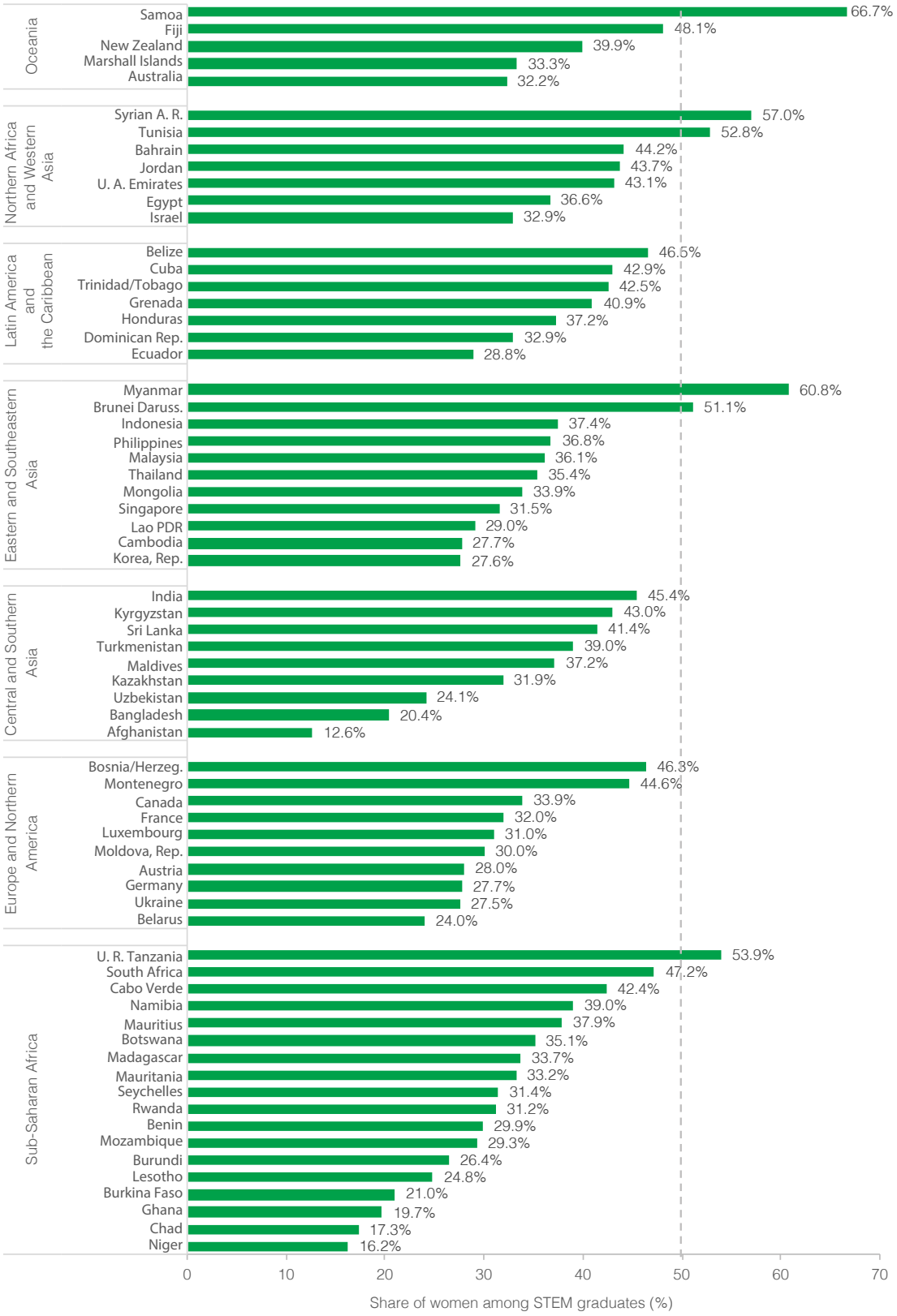
Figure 1. *Gross enrolment ratio in tertiary education, 2010–2021/23*



Notes: Reproduced from Figure 10.5 of UNESCO (2024a).[1] *Source*: UNESCO Institute for Statistics

However, women are less likely to choose science, technology, engineering and mathematics (STEM) fields. In 2018–2023, the global share of STEM graduates who were women was 35%, with variation among countries from 12.6% to 66.7% (Figure 2). In 112 of the 135 countries with data, less than 20% of women graduates had selected STEM degrees (data not shown).

Figure 2. *Share of female STEM graduates, selected countries, 2022 or most recent year*



Notes: Reproduced from Figure 23 of UNESCO (2024b).[2] *Source*: UNESCO Institute for Statistics

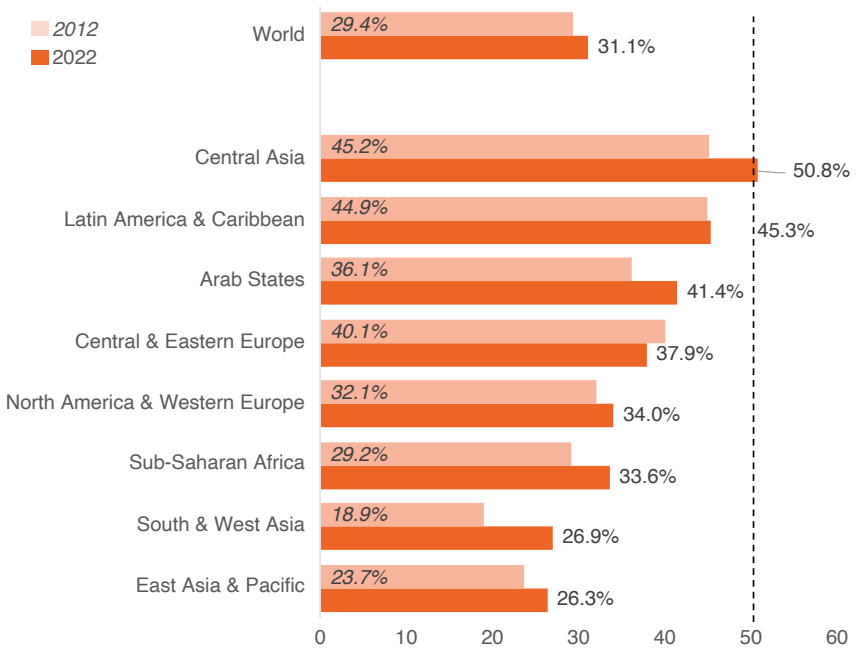
2 JUST ONE IN THREE RESEARCHERS GLOBALLY IS A WOMAN

Women continue to account for a minority of the world's researchers, forming approximately one third: Women represented 31.1% of researchers worldwide in 2022. This estimate uses the most recent values submitted by countries to the unique time series built by the UNESCO Institute for Statistics (UIS) since 1996 (Figure 3).[3]

Global participation of women in the research profession has shown only slight progress over the past decade (2012–2022), increasing from 29.4% to 31.1%. Ongoing efforts to increase gender equity in science have hardly shifted the global value, yet significant disparities remain at national level.

See pages 11-13 for country-level data.

Figure 3. *Share of women among researchers, by region, 2012 and 2022 or most recent year*[4]



Note: Data are based on head counts. Data are not available for all countries; notable absences include the populous countries of Australia, Canada and the USA. Source: UNESCO Institute for Statistics, March 2025

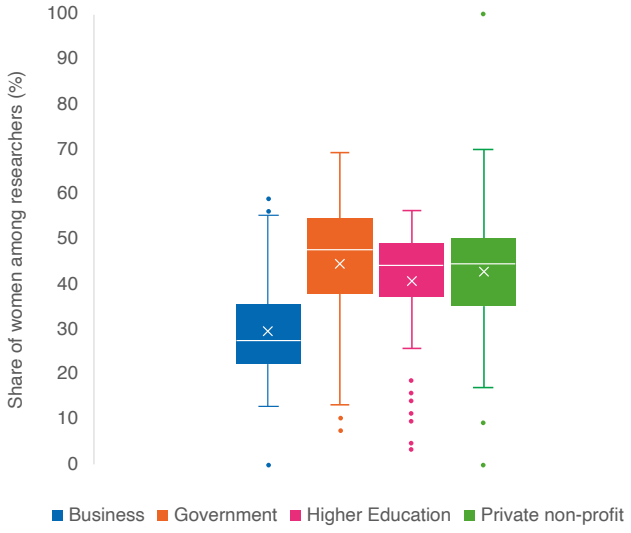
Regional differences are striking.[5] Parity[6] has been reached in Central Asia and in the Latin American and Caribbean region. South and West Asia shows the most impressive growth in the share of women among researchers, from 18.9% in 2012 to 26.9% in 2022, with changes of five percentage points or less in the other regions. The East Asia and Pacific region has the lowest share, yet some of the largest populations of researchers. Central and Eastern Europe is the only region to show a decline in the share of women among researchers, from 40.1% in 2012 to 37.9% in 2022.

These trends in regional and global averages should however be considered as estimates, as they are produced using the full dataset since 1996 and data are missing for several countries.

3 WOMEN'S REPRESENTATION VARIES AMONG SECTORS

To close the gender gap in science, tracking overall participation needs to be supplemented by an understanding of which areas of science women are contributing to and leading in, as well as those from which they remain underrepresented.

Figure 4. *Distribution of women among researchers by sector, 2023 or most recent year*



Notes: In this plot, the centre line indicates the median (central value), the boxes show the interquartile range with the middle 50% of the values, and the 'whiskers' show the full range of values; dots indicate values outside of the range of the most common values (outliers are calculated as the first or third quartile minus or plus, respectively, 1.5 times the interquartile range). Source: UNESCO Institute for Statistics, March 2025

Table 1. *Number of countries where gender parity has been reached, or where women or men are overrepresented among researchers, by sector, 2013–2023*

	Business	Government	Higher education	Private non-profit
Number of countries where women are underrepresented	82	64	74	49
Number of countries with parity	9	34	42	20
Number of countries where women are overrepresented	4	24	10	13

Note: Values are from countries with data for at least one year between 2013 and 2023, using the most recent value. Gender parity is considered as women accounting for 45–55% of the research workforce. Data are not available for all sectors in all countries. Source: UNESCO Institute for Statistics, March 2025

The **business sector**[7] continues to show the lowest gender balance across the science and innovation ecosystem, with a global median of 29.8% (Figure 4). Women are underrepresented in the private sector research workforce in 82 of 95 countries with data for this sector (86%; Table 1). There is substantial variation, with national shares ranging from negligible levels in some countries to a high of 59% in Botswana (Figure 4).[8] Just eight countries have an equitable

proportion of women working as researchers in the business sector, namely Algeria (53.8%), Bosnia and Herzegovina (54.7%), Kazakhstan (53.1%), Kyrgyzstan (53.4%), Mongolia (50%), Mozambique (50%), Namibia (45%) and Uruguay (48.5%). Women are slightly overrepresented in Bolivia (55.6%), Botswana (59.3%), Republic of North Macedonia (56.5%) and Trinidad and Tobago (57.1). Among countries with the largest populations of researchers, the business sector is clearly the furthest from gender parity (Figure 5).

Far more countries are closer to parity in the **higher education**[9] (median: 43.4%), **government**[10] (43.3%) and **private non-profit**[11] sector (40.6%) (Figure 4). However, national values range widely from 0% to 100% women for a given sector, with many of the highest values seen in countries with very small workforces where the hiring of a few people can drastically shift the percentage of women. Approximately one-third of countries have achieved gender parity among researchers in the government and higher education sectors (Table 1).

The presence, or absence, of women in visible leadership roles in higher education is uniquely influential for the aspirations of the next generation of women scientists and scholars. Although women now outnumber men in higher education enrolment in most countries (Figure 2), women are under-represented as researchers and in senior leadership positions.[12] In 2023, just one in four of the world's top 200 higher education institutions were led by women.[13] Representation in academic leadership varies among countries and regions, with particular reason for concern in sub-Saharan Africa and Asia-Pacific.[1] Gender disparities also persist in this research workforce: women are underrepresented among higher education researchers in 72 countries (Table 1).

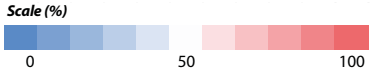
Gender disparities are less marked in the research workforce of the private non-profit sector, but women remain underrepresented in 40 countries (Table 1). Fewer countries report on this relatively smaller segment of the research workforce.

Parity is within reach for all countries, regardless of the size of their workforce: There is no relationship between women's representation and the relative number of researchers in a country (Figure 5). Yet among the top 20 countries in the world by relative number of researchers, only one country has crossed the threshold to gender parity,[5] namely Iceland where women represented 46.8% of researchers in 2022.

Figure 5. *Heat map of the share (%) of women among researchers overall and by sector, selected countries, 2023 or most recent year*

Top 20 countries based on number of researchers per million inhabitants

Country	Researchers per million inhabitants (FTE)	Share of women among researchers (%, HC)	Share of women among researchers by sector (%, HC)			
			Business	Government	Higher education	Private non-profit
Liechtenstein, 2023	18 130	25.4	24.9	30.8	28.0	
Korea, Rep., 2022	9 435	23.0	19.2	28.7	35.2	31.3
Denmark, 2019	8 736	35.3	25.5	50.2	44.5	62.8
Sweden, 2021	8 624	34.6	25.6	50.3	46.6	46.2
Finland, 2022	8 073	33.6	18.4	44.2	50.0	57.6
Singapore, 2021	7 917	31.8	27.3	37.1	36.0	
Norway, 2022	7 395	38.7	22.8	50.9	52.0	
Belgium, 2021	6 983	33.5	27.7	36.1	43.4	53.4
Iceland, 2022	6 864	46.8	34.8	60.7	54.5	
Austria, 2021	6 659	31.3	18.7	43.0	42.8	49.2
Netherlands, 2022	6 319	30.2	24.5	45.8	46.3	
Switzerland, 2022	6 021	37.5	29.7	41.2	41.6	
Germany, 2021	5 787	29.4	15.6	38.3	41.4	
Portugal, 2022	5 687	42.1	29.8	61.4	50.5	49.8
Japan, 2022	5 630	18.3	12.2	21.3	28.9	17.3
Ireland, 2021	5 512	37.4	24.5	49.6	48.2	
Slovenia, 2022	5 370	35.0	24.8	49.1	44.7	45.3
France, 2022	5 182	29.7	22.6	38.7	42.0	44.7
Greece, 2022	4 927	40.1	31.4	43.2	43.4	55.6
Luxembourg, 2021	4 879	29.9	14.5	38.0	36.8	



Note: Researchers per million inhabitants is an official indicator for Sustainable Development Goal 9; see https://uis.unesco.org/en/topic/sustainable-development-goal-9-5. Canada ranks 17 by researchers per million inhabitants but lacks data for the share of women among researchers; for that reason, Greece is included. FTE: full time equivalents; HC: head counts. Source: UNESCO Institute for Statistics, March 2025

Women in Quantum Science

The United Nations General Assembly proclaimed 2025 as the International Year of Quantum Science and Technology. As the lead agency, UNESCO is committed to ensuring quantum advancements serve all of humanity. As quantum technologies rapidly evolve, significant disparities in research capacity, technological adoption and education access persist, particularly across developing regions. These disparities are not only regional—they are also gendered.

The talent pipeline into the quantum workforce shows severe disparities. According to recruitment data from Quantum Futures, reported by the London School of Economics and Political Science, fewer than 2% of applicants for jobs in the quantum sector are women—just one in 54. Furthermore, 80% of quantum companies reportedly lack any senior female figures.[14]

In Europe, the QuantERA network of research funders supports the development of quantum technologies. Across its first three calls (2017, 2019 and 2021), women researchers acted as principal investigators in 51 of 400 national research teams, and just nine women were among the 77 leaders of international research consortia.[15]

INTERNATIONAL YEAR OF
Quantum Science and Technology

Yet this underrepresentation is not due to lack of interest. At university level, women enrolled in physical sciences and computing disciplines report comparable enthusiasm for quantum science as their male peers.[16] To fully harness this interest, especially among women, it is important to foster cross-disciplinary pathways because quantum sciences rely on many diverse fields.

Calls for change are growing: A group of physics professors initiated the Women for Quantum (W4Q) Manifesto of Values, with more than 150 signatories at the time of writing, advocating for a more inclusive future in quantum.[17]

To inform global efforts, UNESCO has partnered with IBM and the Unitary Foundation to launch a *Global Survey on the Research Status and Infrastructure in Quantum Science and Technology*. Preliminary findings show that a third of respondents work around mostly male researchers in quantum sciences at their institution, a quarter reported gender balance, and another third said they did not know or data were not available regarding gender distribution at their institution. Reported challenges facing women in the field include, among others, a lack of role models or mentors, challenges in achieving work-life balance including a lack of family-friendly policies, and limited representation of women in leadership positions. Respondents highlighted needed actions to improve gender equity, ranging from mentorship and recognition to targeted resourcing and inclusive policies. The full report is forthcoming at: https://www.unesco.org/en/years/quantum-science-technology

Women in Water Science

The year 2025 marks a dual milestone: 50 years of the Intergovernmental Hydrological Programme and 60 years of UNESCO's commitment to water sciences. These anniversaries offer an opportunity to reflect not only on scientific progress but also on who has contributed to that progress—and who has been left out.

Despite the increasing relevance of gender in water management and policy, past and contemporary research is largely missing information about impact on women, the contributions of women or gender considerations for the interfaces of society and water. A bibliometric study found that out of thousands of scientific articles published in water-related disciplines, fewer than 100 mentioned both gender, or women, and hydrology.[18] In 2022, nine of the top 21 hydrology journals published no articles referencing gender, while the remainder published between one and 14.[18]

This absence in the literature or data does not reflect an absence of women in the field of water sciences. On the contrary, women have contributed to water science throughout history, and today they represent a growing share of national hydrological society memberships. Representation in leadership is also gradually improving.[18] However, the field still displays structural barriers to success. In the 21 leading hydrology journals, women occupied just 4% to 38% of editorial board seats, with a median of 18%.[18] Editorial board composition is positively linked to publication success for women, particularly for lead and senior authors.[19]

Equity in workforce and pay also remains elusive, albeit with limited data. Women constitute less than 17% of the total paid workforce in the water sector, with even fewer in research and decision-making positions.[20]

UNESCO is working to accelerate progress towards gender equality in the water sector and to give more visibility to women researchers in water sciences. On behalf of the UN system, UNESCO is coordinating the United Nations World Water Development Report 2026 on water and gender.[21] Another publication in the UNESCO *Global Water Security Issues* series—titled "*Water and Gender: Towards Women's Equal Rights and Opportunities*"—is in preparation for 2026, with 68% of proposals submitted by women lead authors, up from 45% in the fourth (2023) and 50% in the fifth (2025) volumes. The book *Emerging Pollutants: Protecting Water Quality for Health and Environment*, co-published with Springer Nature in 2025, features pioneering research by women scientists from around the world with 13 of 21 chapters led by women. Furthermore, the UNESCO World Water Assessment Programme implements a water and gender programme, including methodologies and tools to collect sex-disaggregated data.[22]



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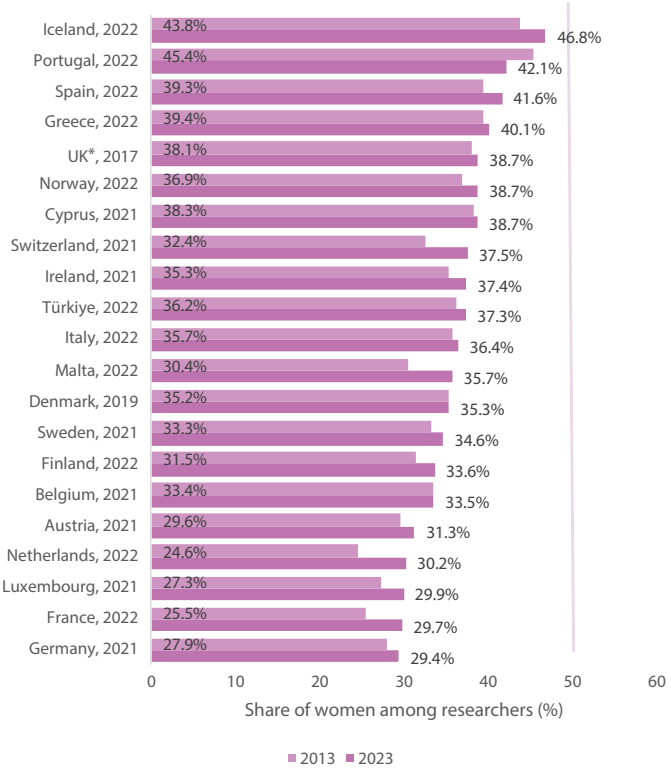
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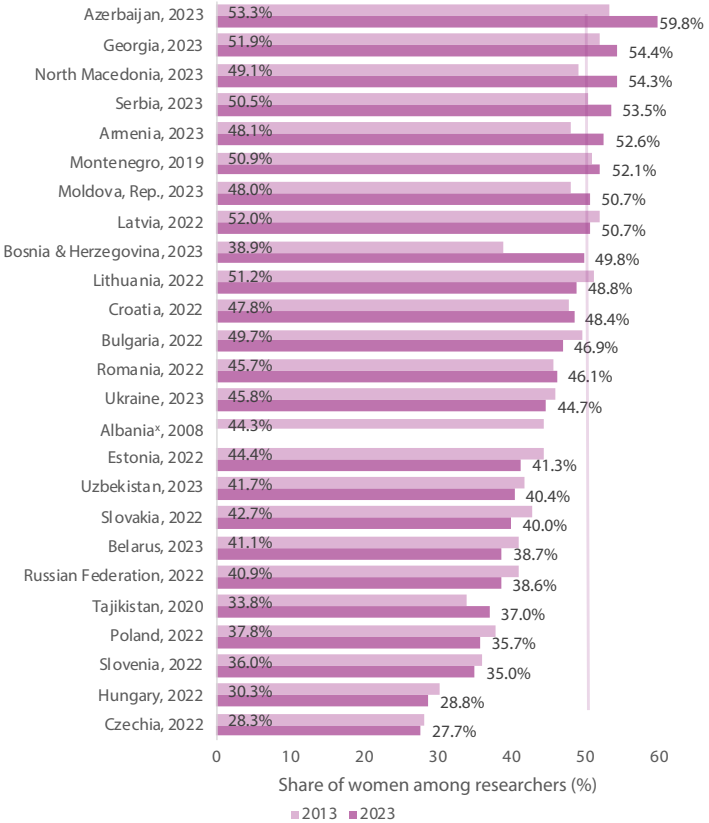
Country-level data

Figure 6. *Women in research in Western Europe*
Female researchers as a percentage of total researchers (head counts), 2013 and 2023 or latest year available



Notes: Since 1996, data are available for 21 of the 22 countries in Western Europe. No internationally comparable data are available for North American countries.
*Data predate 2019; ˣData predate 2014.
Data source: UNESCO Institute for Statistics, March 2025

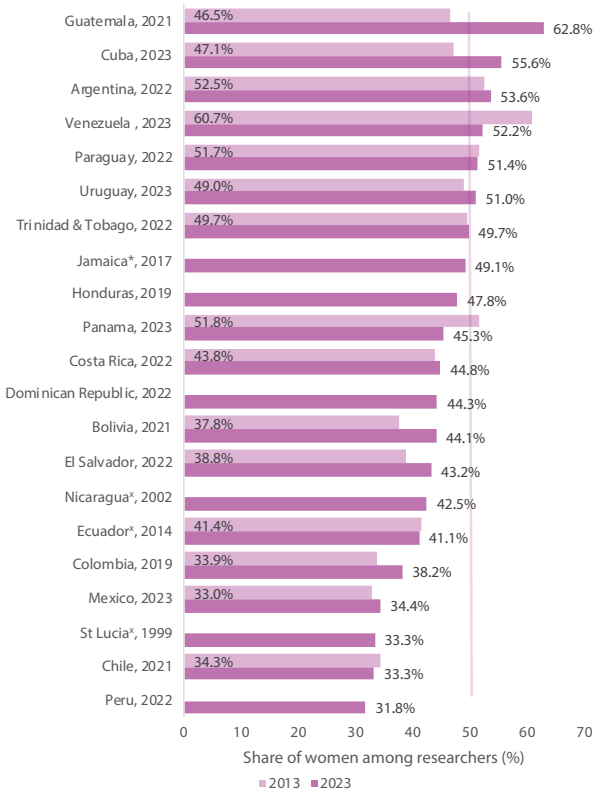
Figure 7. *Women in research in Eastern Europe*
Female researchers as a percentage of total researchers (head counts), 2013 and 2023 or latest year available



Notes: Since 1996, data are available for all of the 25 countries in Eastern Europe.
*Data predate 2019; ˣData predate 2014.
Data source: UNESCO Institute for Statistics, March 2025

Figure 8. *Women in research in Latin America and the Caribbean*

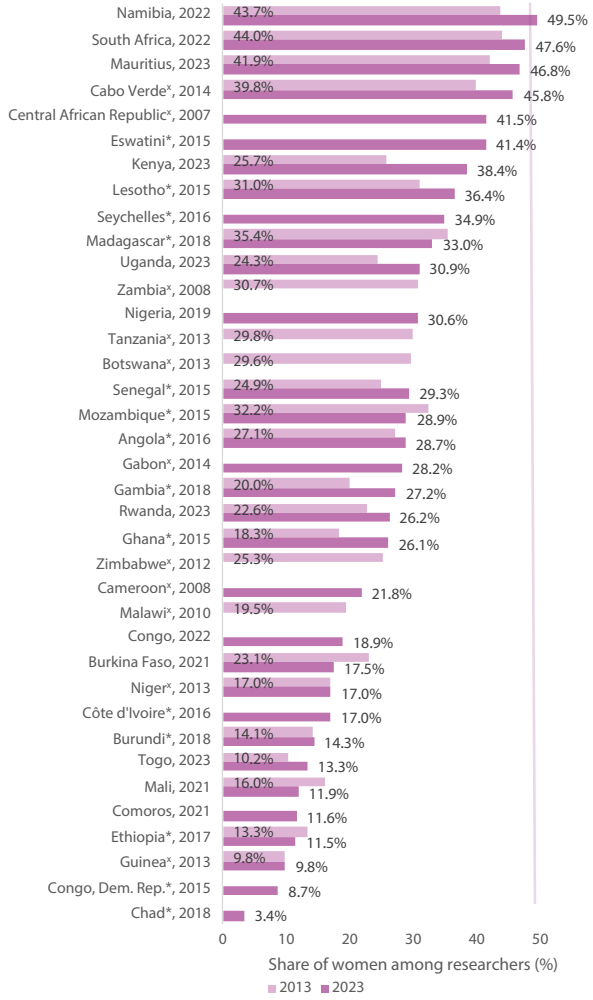
Female researchers as a percentage of total researchers (head counts), 2013 and 2023 or latest year available



Notes: Since 1996, data are available for 16 of 22 countries in Latin America and 5 of 13 countries in the Caribbean. *Data predate 2019; [x]Data predate 2014. Data source: UNESCO Institute for Statistics, March 2025

Figure 9. *Women in research in sub-Saharan Africa*

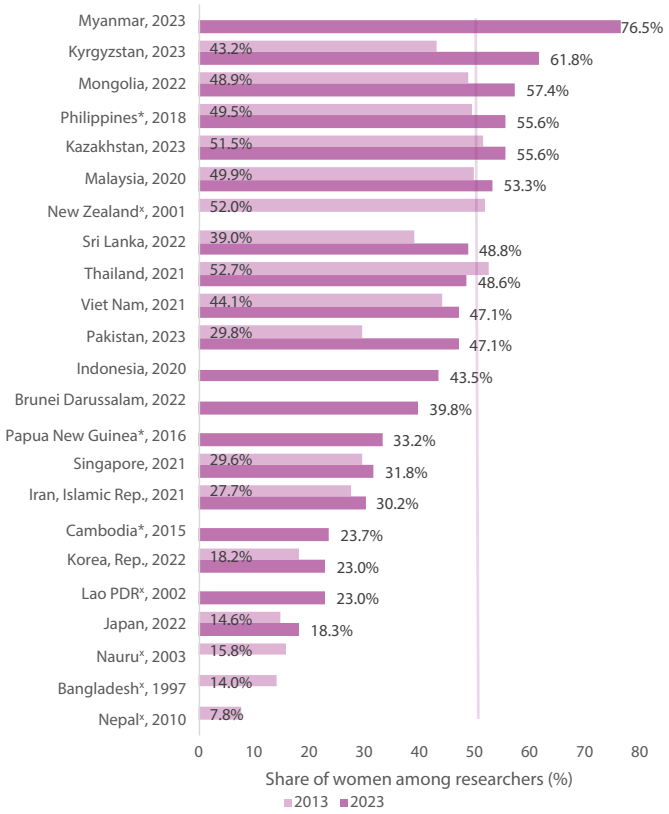
Female researchers as a percentage of total researchers (head counts), 2013 and 2023 or latest year available



Notes: Since 1996, data are available for 37 of the 47 countries in sub-Saharan Africa. *Data predate 2019; [x]Data predate 2014. Data source: UNESCO Institute for Statistics, March 2025

Figure 10. *Women in research in Asia and the Pacific*

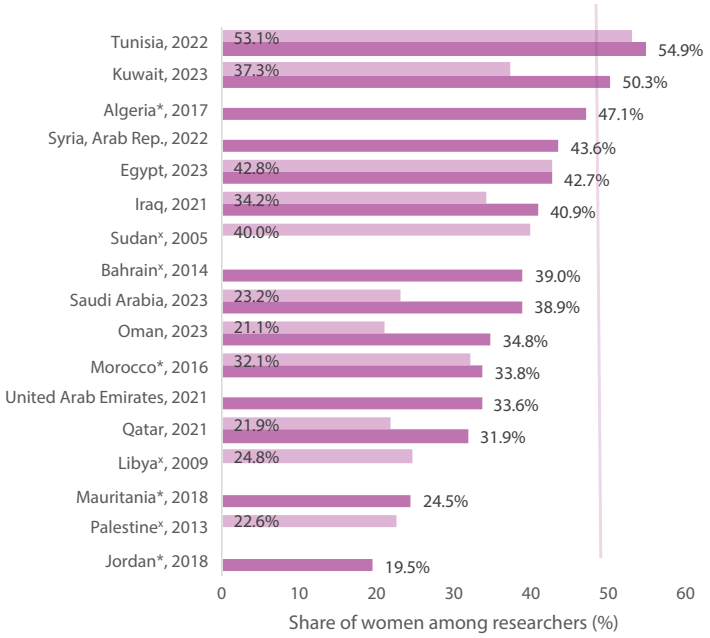
Female researchers as a percentage of total researchers (head counts), 2013 and 2023 or latest year available



Notes: Since 1996, data are available for 23 of the 44 countries in the Asia-Pacific region. *Data predate 2019; ˣData predate 2014. Data source: UNESCO Institute for Statistics, March 2025

Figure 11. *Women in research in Arab States*

Female researchers as a percentage of total researchers (head counts), 2013 and 2023 or latest year available



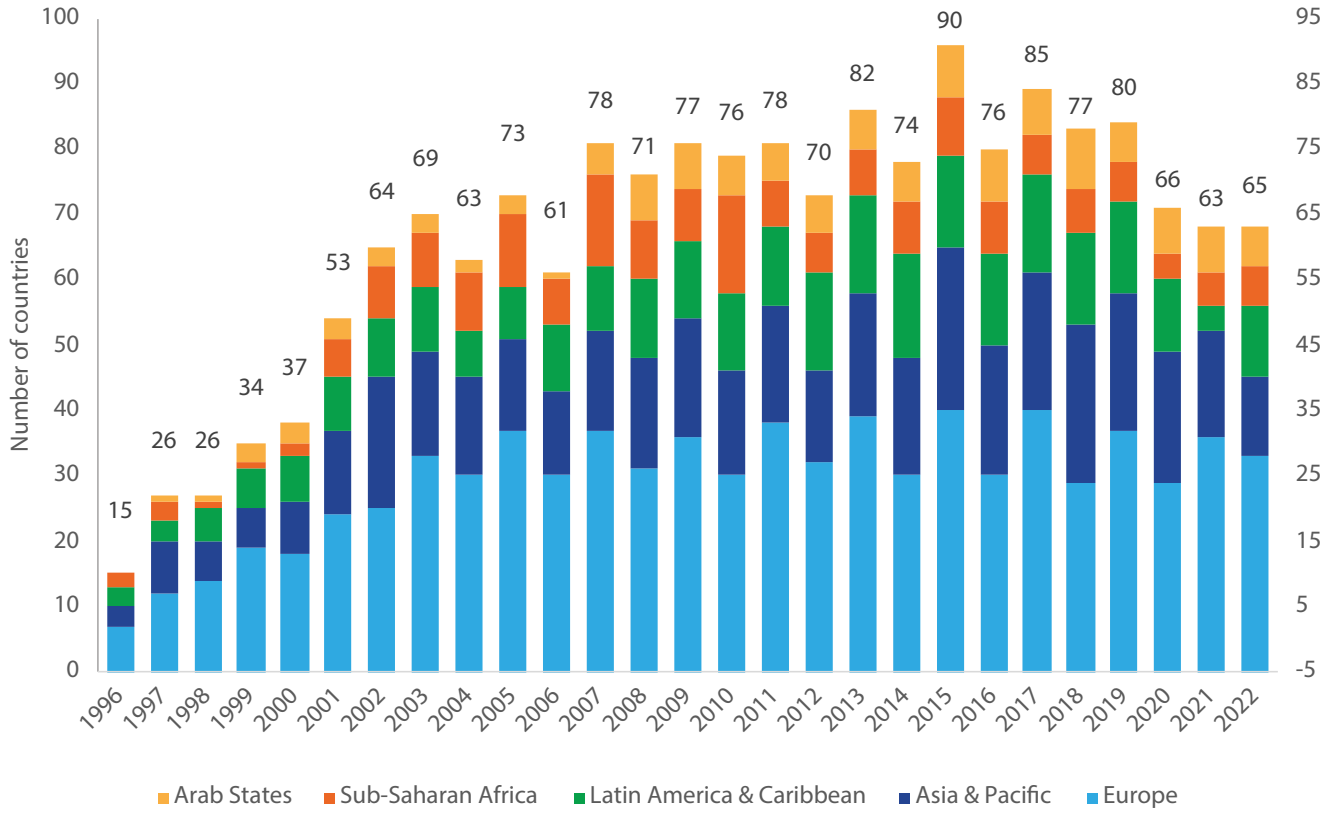
Notes: Since 1996, data are available for 17 of the 19 Arab States. *Data predate 2019; ˣData predate 2014. Data source: UNESCO Institute for Statistics, March 2025

4. Assessments demand data

Between 2012 and 2022, 131 countries [23] reported the share of women researchers in national science, technology and innovation systems.

Fewer than 90 countries report on the share of women among researchers each year—less than half of all countries (Figure 12). Although reporting had increased to 78 countries on average reporting annually between 2007 and 2018, fewer than 66 countries have reported each year since 2019.

Figure 12. *Number of countries reporting on the share of women among researchers in their science systems, 1996–2022*



Notes: Restricted to 2022 because only 30 countries have submitted data for 2023 as of March 2025. Data source: UNESCO Institute for Statistics, March 2025

Since 1996, 49 countries have never reported internationally comparable data on women's representation in their scientific workforce.[24]

A total of 163 countries have reported on their overall scientific workforce at least once since 1996, yet only 131 countries have reported on the representation of women in science since 2012 and 103 countries since 2018. More than half of countries in the Asia-Pacific region (see Figure 10) and over one-third of countries in the Americas and Caribbean (see Figure 8) have not reported data regarding women among researchers since 2012.

Simply counting the number of people who conduct research at least a part of their professional time does not convey the full situation for women scientists. Numerical parity is used as a proxy indicator for the larger socio-cultural context in which women researchers operate.

5. Closing the gender gap

Closing the gender gap in science requires sustained, collective commitment. UNESCO's *Call to Action to Close the Gender Gap in Science*, launched in February 2024, invites bold steps to remove systemic barriers and accelerate gender equality in science. The Call outlines priority actions for dismantling gender stereotypes and biases in science, opening educational pathways for girls in science and creating workplace environments that attract, retain and advance women scientists.

This factsheet is part of a growing knowledge base to support the Call's implementation. Readers are encouraged to explore the tools, guidance and examples of good practice or to join one of UNESCO's global Working Groups: https://www.unesco.org/en/science-technology-and-innovation/cta

How to get involved



Please consult the UNESCO Institute for Statistics website http://uis.unesco.org for methodology and access the database at https://databrowser.uis.unesco.org/

A note regarding data coverage:
For some countries, reported data do not cover all sectors. Data may be partial or provisional and may not correspond exactly to Frascati Manual recommendations.

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Graphic design and layout: Claudia Tortello

Supported by the L'Oreal—UNESCO For Women In Science programme with financial contribution from the Fondation L'Oréal

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Notes:

1 UNESCO. 2024a. *Global Education Monitoring Report 2024/5: Leadership in Education – Lead for learning*. Paris: UNESCO.

2 UNESCO. 2024b. *Global Education Monitoring Report: Gender report – Technology on her terms*. Paris: UNESCO.

3 Countries report on female researchers or research and development personnel in headcounts or full-time equivalents. Headcounts indicate the total number of persons employed in research and development. The headcount value includes staff employed both full-time and part-time. The full-time equivalent (FTE) of personnel is defined as the ratio of working hours actually spent on research and development during a specific reference period (usually a calendar year) divided by the total number of hours conventionally worked in the same period by an individual or by a group. In other words, one full-time equivalent may be thought of as one person-year.

4 The regional averages are based on available data for 155 countries and are derived using the nearest year's data when 2012 or 2022 data are unavailable. For 151 countries, estimates are based on data for researchers in headcounts. For the remaining countries, estimates were derived from other available data: researchers in full-time equivalents (India), total R&D personnel (China) and UIS estimations (Brazil and Israel). Data for other countries, including Australia, Canada and the United States of America, are not available. Only four Caribbean countries have data. Regions in this graph follow the UIS regional groupings.

5 There is no regional value for North America because Canada and the USA do not report data on the share of female researchers. Mexico is included in the Latin America and Caribbean region here.

6 Gender parity is considered as women accounting for 45–55% of the research workforce.

7 In the context of research and development (R&D) statistics, the Business Enterprise sector comprises: • All resident corporations, including not only legally incorporated enterprises, regardless of the residence of their shareholders. This group includes all other types of quasi-corporations, i.e. units capable of generating a profit or other financial gain for their owners, recognised by law as separate legal entities from their owners, and set up for purposes of engaging in market production at prices that are economically significant. • The unincorporated branches of non-resident enterprises are deemed to be resident because they are engaged in production on the economic territory on a long-term basis. • All resident non-profit institutions (NPIs) that are market producers of goods or services or serve business. • This sector comprises both private and public enterprises.

8 Based on countries reporting data on the share of women among researchers by sector between 2013 and 2023.

9 In the context of R&D statistics, the Higher Education sector comprises: • All universities, colleges of technology and other institutions providing formal tertiary education programmes (i.e. ISCED levels 5, 6, 7, or 8), whatever their source of finance or legal status. • All research institutes, centres, experimental stations and clinics that have their R&D activities under the direct control of, or are administered by, tertiary education institutions.

10 In the context of R&D statistics, the Government sector consists of the following groups of resident institutional units: • All units of central (federal), regional (state) or local (municipal) government, including social security funds, except those units that provide higher education services or fit the description of higher education institutions (see the definition of Higher education sector). • All non-market NPIs that are controlled by government units that are not part of the Higher education sector. • This sector does not include public corporations, even when all the equity of such corporations is owned by government units. Public enterprises are included in the Business enterprise sector.

11 In the context of R&D statistics, the Private Non-profit sector comprises: • All non-profit institutions serving households (NPISH), except those classified as part of the Higher education sector. • Households and private individuals engaged or not engaged in market activities.

12 Gandhi and Sen. 2021. cited in UNESCO. 2024. *Global Education Monitoring Report 2024/5: Leadership in Education*. Paris: UNESCO.

13 Times Higher Education 2023, https://www.timeshighereducation.com/world-university-rankings/women-lead-quarter-top-200-universities; see for open access summary: https://www.weforum.org/stories/2023/03/top-university-rankings-women-leadership/

14 See: https://www.lse.ac.uk/News/Latest-news-from-LSE/2023/c-March-2023/More-diversity-and-inclusion-needed-quantum-computing

15 See: https://quantera.eu/women-leaders-in-quantum-research/

16 Rosenberg, J.L., Holincheck N. and Colandene M. 2024. Science, technology, engineering, and mathematics undergraduates' knowledge and interest in quantum careers: Barriers and opportunities to building a diverse quantum workforce. *Physical Review Physics Education Research* 20:010138. https://doi.org/10.1103/PhysRevPhysEducRes.20.010138

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23 Between 2012 and 2022, 131 countries, including Liechtenstein, reported standardized data for at least one year on the share of female researchers in head counts; an additional two countries (China and Israel) reported data for all R&D personnel and one country (India) reported data in full-time equivalents.

24 An additional country (China) reported data for all R&D personnel and one country (India) reported data in full-time equivalents.


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